








Statistics of Financial Markets

Chapter 4: Probability



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Acronyms used in this material

- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model)
- **ARCH**: AutoRegressive Conditional Heteroskedasticity
- **BET**: Bucharest Exchange Trading (index)
- **CDF**: Cumulative Distribution Function
- **CLT**: Central Limit Theorem
- **CRR**: Cox–Ross–Rubinstein (binomial model)
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GBM**: Geometric Brownian Motion
- **IBM**: International Business Machines (company)
- **LCG**: Linear Congruential Generator
- **LLN**: Law of Large Numbers
- **PCG64**: Permuted Congruential Generator, 64-bit (NumPy default)
- **PMF**: Probability Mass Function
- **RANDU**: RANDU (IBM random number generator of the 1960s)
- **SFE**: Statistics of Financial Markets, Exercises (Quantlet collection of the textbook)

Today's Question, Route and Learning Outcomes

- **Question**: which probability tools do we need to describe returns and to simulate prices?
  - every later chapter (tails, volatility, VaR, efficiency) is built on them
- **Route and outcomes**: after this chapter you can
  - compute expectations, variances, covariances and correlations, and the volatility of a portfolio
  - explain why uncorrelated returns can still be dependent, and measure it
  - use conditional expectation and conditional variance, the building blocks of GARCH
  - generate random numbers with the inverse transform and estimate a probability by Monte Carlo, with its standard error
  - define and simulate the binomial model, a random walk, a martingale, white noise, AR(1), the Wiener process and GBM
- Real data, 2000–2026: S&P 500, DAX, BET and Bitcoin (from 2014)

Reading and Tools

- Textbook: Franke, Härdle & Hafner (2019), *Statistics of Financial Markets*, 5th ed., Ch. 3 (Sec. 3.1–3.5) and Ch. 4 (discrete-time stochastic processes)
  - exercises with solutions: Borak, Härdle & López-Cabrera (2013), Ch. 3–4
- Monte Carlo methods in finance: Glasserman (2003)
- Python Quantlets of this chapter: Quantlets/Ch_04
  - ported from the SFE Quantlets of the textbook (SFErandu, SFErangen1, SFErangen2, SFEBinomp, SFEWienerProcess, SFEsimGBM)
  - each chart in these slides links to the Quantlet that draws it
- Lecture notebook: open in Google Colab
- Video course: Statistics of Financial Markets

1654: the Problem of Points

- Two players stop a fair game before the end: how should the stake be split?
- Blaise Pascal and Pierre de Fermat solved it in letters (1654): split by the probability of winning, not by the score
- Example: A needs 1 more win, B needs 2; at most two more rounds
  - B wins only if B wins both rounds: probability $1/4$
  - A should get 75% of the stake: its **expected** gain
- The same idea prices a derivative today: a fair price is an expected payoff

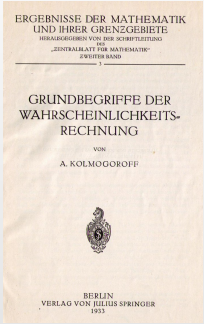


Blaise Pascal (1623–1662)

Portrait: unknown author, after François II Quesnel (c. 1690); public domain; Wikimedia Commons

Kolmogorov's Axioms (1933)

- A **probability space** $(\Omega, \mathcal{F}, P)$ Kolmogoroff (1933)
  - Ω: all possible outcomes (all possible paths of tomorrow's market)
  - $\mathcal{F}$: the events we can assign a probability to (a σ-algebra)
  - P: a function from events to $[0, 1]$
- Axioms, for any event $A \in \mathcal{F}$
  - $P(A) \geq 0$: no probability is negative
  - $P(\Omega) = 1$: some outcome certainly occurs
  - $P(\cup_i A_i) = \sum_i P(A_i)$ for disjoint $A_1, A_2, \ldots$ (no two can occur together): the probability of "one of them" is the sum
- Everything else (expectation, independence, processes) is built on these three rules


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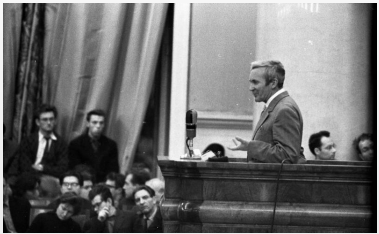
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1933

Title page of the *Grundbegriffe*, Springer 1933

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Andrey Kolmogorov

- Andrey Nikolaevich Kolmogorov (1903–1987), Moscow State University
- Made probability a branch of measure theory: events are sets, probability is a measure
- Also: conditional expectation in its modern form, Markov processes, turbulence, complexity
- The Kolmogorov–Smirnov test (Chapter 6) carries his name



Kolmogorov lecturing in Moscow, 1963–1964

Photo: Vsevolod Tarasevich (1963–1964); CC BY 4.0; Wikimedia Commons

Events, Conditional Probability, Independence

- **Conditional probability**: $P(A \mid B) = P(A \cap B)/P(B)$, for $P(B) > 0$
  - A, B: two events; $A \cap B$: both occur
  - in words: among the cases in which B occurs, the share in which A also occurs
- **Bayes' rule**: $P(B \mid A) = P(A \mid B)P(B)/P(A)$
  - $P(B)$: the probability of B before we learn A; $P(B \mid A)$: the updated probability once A is observed
  - it reverses the direction of conditioning: from "signal given crisis" to "crisis given signal"
- **Independent** events: $P(A \cap B) = P(A)P(B)$, i.e. $P(A \mid B) = P(A)$
  - learning that B occurred does not change the probability of A

Example: Down Days in a Row

- S&P 500, 2000–2026: $P(\text{down day}) = 46.3\%$
  - after a down day: $P(\text{down} \mid \text{down yesterday}) = 43.4\%$
  - two down days in a row: 20.1% in the data
  - under independence: $P(\text{down})^2 = 21.4\%$, close to the data
- BET: 50.0% after a down day vs 42.6% after an up day
  - the sign of the return persists from one day to the next
  - a consequence of thin trading: prices adjust to news over several days

Recap: From Games to Axioms

- Probability space $(\Omega, \mathcal{F}, P)$ and three axioms
- Fair value = expected value: the idea of Pascal and Fermat
- Conditional probability measures how information changes the odds

Random Variables

- A **random variable** $X : \Omega \to \mathbb{R}$ assigns a number to each outcome
  - tomorrow's log return, the number of up days in a week, the loss of a portfolio
- **Discrete**: finitely or countably many values x_i
  - **PMF** (probability mass function): $p(x_i) = P(X = x_i)$, with $\sum_i p(x_i) = 1$
  - $p(x_i)$: the probability of the value x_i; the probabilities of all values add up to 1
- **Continuous**: $P(a < X \leq b) = \int_a^b f(x)\,dx$
  - the probability of an interval $(a, b]$ is the area under f between a and b
  - f is the **PDF** (probability density function): $f \geq 0$ and the total area $\int f = 1$
  - a single point has no area: $P(X = x) = 0$ for every x
- Textbook: Franke, Härdle & Hafner (2019), Sec. 3.1

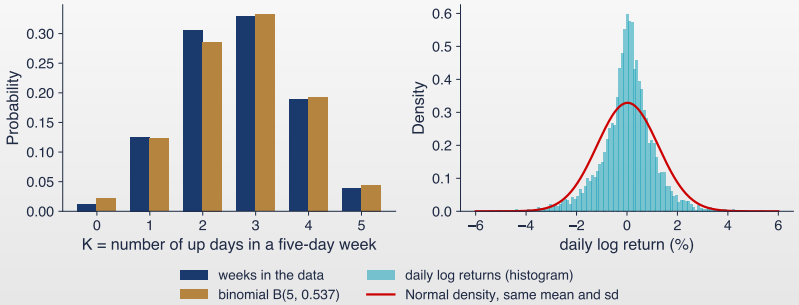
The Distribution Function and the Quantile Function

- **CDF** (cumulative distribution function): $F(x) = P(X \leq x)$
  - the probability that X does not exceed the threshold x
  - non-decreasing, right-continuous, from $F(-\infty) = 0$ to $F(\infty) = 1$
  - for a continuous X the density is its derivative: $f = F'$
- **Quantile** of order α: $q_\alpha = F^{-1}(\alpha) = \inf\{x : F(x) \geq \alpha\}$
  - $\alpha \in (0, 1)$: a probability; q_α is the smallest x with $P(X \leq x) \geq \alpha$
  - F^{-1}: the quantile function, the inverse of the CDF; inf: the smallest value of the set
  - median $= q_{0.5}$; the 1% quantile is the return exceeded on 99% of the days

Value at Risk and the Empirical CDF

- **VaR** (Value at Risk) 1%: $\text{VaR}_{1\%} = -q_{0.01}$
  - $q_{0.01}$: the 1% quantile of the returns, a negative number
  - the minus sign turns it into a loss: the loss exceeded with probability 1%
  - estimation and backtesting in Chapter 10
- The **empirical CDF** of a sample: $\hat{F}(x) = \frac{1}{n}\#\{t : r_t \leq x\}$
  - $r_1, \ldots, r_n$: the observed returns; n: the sample size
  - $\#\{\cdot\}$: the number of days in the set; $\hat{F}(x)$ is the share of days with a return at most x
  - the hat marks an estimate computed from data

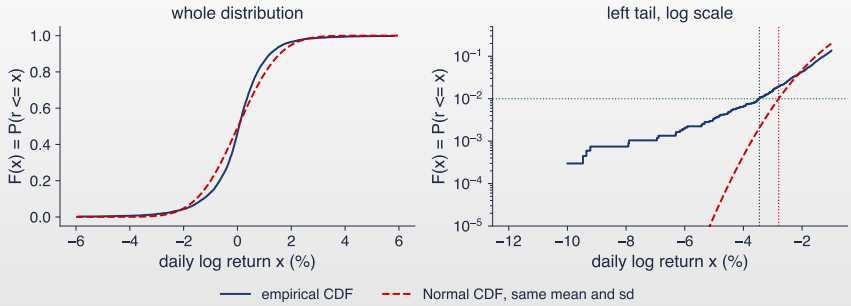
A Discrete and a Continuous Random Variable



- Left: K = number of up days in a full week of the S&P 500 (1 144 weeks)
  - compared with $B(5, p)$, the binomial distribution: 5 days, each up with probability $p = 0.537$
  - mean 2.67 vs 2.69; variance 1.16 vs 1.24
- Right: daily log returns, a continuous random variable; the histogram estimates the PDF
- The binomial fits K well; the Normal density misses the peak and the tails of the returns (Chapter 2)

SFM_ch4_random_variables

Empirical CDF and the 1% Quantile of the S&P 500



- $n = 6\,714$ days; median 0.063%; $\hat{F}(0) = 46.3\%$ of the days are not up days
- VaR 1%: empirical 3.45%, Normal 2.80%; $P(r < -3\%)$: 1.55% vs 0.63%
- $P(r < -5\%)$: 0.34% in the data, 199 times the value under the Normal distribution (1.7×10^{-3}%)

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Recap: Random Variables

- PMF for discrete, PDF for continuous variables; the CDF works for both
- Quantiles invert the CDF; VaR 1% is minus the 1% quantile
- The tails of the empirical CDF are much heavier than the Normal ones

Expectation

- **Expectation** (mean): $E[X] = \sum_i x_i p(x_i)$ or $E[X] = \int x f(x)\, dx$
  - x_i, $p(x_i)$: the values and their probabilities; f: the density
  - the centre of mass of the distribution; it exists only if $E|X| < \infty$ (not for Cauchy, Chapter 3)
- Of a function: $E[g(X)] = \int g(x) f(x)\, dx$, without finding the distribution of $g(X)$
- **Linearity**: $E[aX + bY + c] = aE[X] + bE[Y] + c$, always (no independence needed)
  - a, b, c: constants; X, Y: any two random variables
  - the expected return of a portfolio is the weighted mean of the expected returns
- In general $E[g(X)] \neq g(E[X])$
  - Jensen: for a convex g, $E[g(X)] \geq g(E[X])$; for example $E[e^X] > e^{E[X]}$
  - the source of the volatility drag of Chapter 1

Variance and Moments

- **Variance**: $\mathrm{Var}(X) = E[(X-\mu)^2] = E[X^2] - \mu^2$
  - $\mu = E[X]$; the variance is the mean squared distance from the mean
  - standard deviation $\sigma = \sqrt{\mathrm{Var}(X)}$, in the units of X; for returns it is the volatility
- $\mathrm{Var}(aX + b) = a^2 \mathrm{Var}(X)$
  - adding a constant b does not change risk; multiplying by a multiplies the variance by a^2
- Moments: $E[X^k]$ for $k = 1, 2, \ldots$
  - skewness $E[(X-\mu)^3]/\sigma^3$: the asymmetry of the distribution (0 for the Normal distribution)
  - kurtosis $E[(X-\mu)^4]/\sigma^4$: equals 3 for the Normal distribution
  - excess kurtosis = kurtosis -3; positive for heavy tails (Chapter 2)

Worked Example: Annual Mean and Volatility

- S&P 500, daily log returns r_t
  - daily mean 0.0247%, daily standard deviation $\sigma = 1.213\%$
  - $A = 252$: the number of trading days per year
- Assume i.i.d. daily returns (independent and identically distributed)
  - the annual log return is the sum of the A daily ones: $\sum_{t=1}^{A} r_t$
- Mean: $E[\sum r_t] = A \times 0.0247 = 6.2\%$ a year
  - linearity of expectation
- Variance: $\mathrm{Var}(\sum r_t) = A\sigma^2$, because the covariances are zero
  - annual volatility $\sigma\sqrt{A} = 1.213\sqrt{252} = 19.3\%$
  - the square-root-of-time rule: volatility grows with $\sqrt{h}$ over h periods

How Rare Is a 4-Sigma Day? Chebyshev's Bound

- **Chebyshev's inequality**: for any X with finite variance, $P(|X - \mu| \geq k\sigma) \leq 1/k^2$
  - μ, σ: the mean and the standard deviation; $k > 1$: the distance from the mean, in standard deviations
  - it uses only the mean and the variance, so it holds for every distribution
- $k = 4$, S&P 500, 2000–2026
  - Chebyshev bound: at most 6.25% of the days
  - Normal distribution: 0.0063%
  - data: 46 days, 0.69%
- The data lie between the two
  - the Normal distribution is far too optimistic; the worst case of Chebyshev is too pessimistic

Recap: Expectation and Variance

- Expectation is linear; variance scales with the square
- For i.i.d. returns the mean grows with h and the volatility with $\sqrt{h}$
- Chebyshev bounds tail probabilities for any distribution

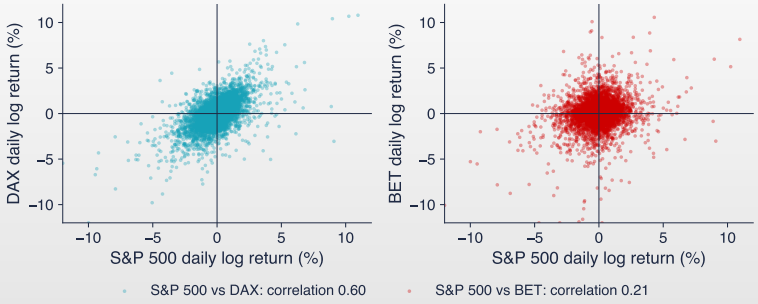
Joint Distribution and Marginals

- **Joint CDF**: $F(x, y) = P(X \leq x, Y \leq y)$
  - the probability that both variables are below their thresholds at the same time
  - for two returns: both markets fall below given levels on the same day
- **Marginal** CDF: $F_X(x) = F(x, \infty)$
  - the distribution of X alone, whatever the value of Y
- **Joint PDF** $f(x, y)$: $P((X, Y) \in B) = \iint_B f(x, y)\, dx\, dy$
  - B: a region of the plane; the probability is the volume under f above B
  - marginal density: $f_X(x) = \int f(x, y)\, dy$, integrating Y out

Covariance, Correlation and the Variance of a Sum

- **Covariance**: $\mathrm{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = E[XY] - \mu_X \mu_Y$
  - μ_X, μ_Y: the means; positive when X and Y tend to be above their means together
- **Correlation**: $\rho = \mathrm{Cov}(X, Y)/(\sigma_X \sigma_Y) \in [-1, 1]$
  - the covariance divided by the two standard deviations: a number without units
  - measures **linear** dependence only; $|\rho| = 1$ iff $Y = a + bX$
- **Variance of a sum**: $\mathrm{Var}(aX + bY) = a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\,\mathrm{Cov}(X, Y)$
  - a, b: constants (in a portfolio, the weights); the covariance term can raise or lower the risk
  - many assets: $\sigma_p^2 = w^\top \Sigma w$, with w the vector of weights and Σ the covariance matrix
  - σ_p^2: the variance of the portfolio; $w^\top$: the transposed vector

Joint Distributions of Daily Returns



- Common trading days only: prices are joined first, then differenced ($n = 6\,605$ days for S&P 500 and DAX)
- S&P 500 and DAX: $\rho = 0.60$; S&P 500 and BET: $\rho = 0.21$
- The cloud is elliptic in the centre, but extreme days are joint: crises hit all markets together

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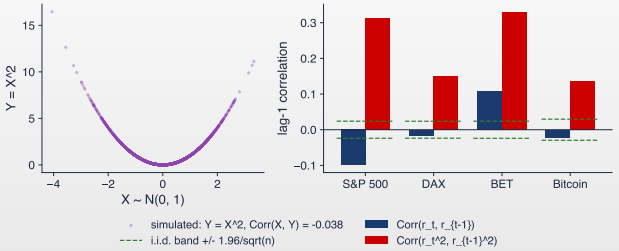
Worked Example: a 50/50 Portfolio of S&P 500 and DAX

- Daily standard deviations: $\sigma_1 = 1.22\%$ (S&P 500), $\sigma_2 = 1.43\%$ (DAX)
  - covariance 1.045; correlation 0.60
  - weights $a = b = 0.5$, so $a^2 = b^2 = 0.25$ and $2ab = 0.5$
- $\sigma_p^2 = 0.25\sigma_1^2 + 0.25\sigma_2^2 + 2 \times 0.25\,\mathrm{Cov} = 0.883 + 0.522 = 1.405$
  - $\sigma_p = 1.19\%$, below the average volatility 1.32%: diversification
- S&P 500 and BET, lower correlation: $\sigma_p = 1.04\%$ vs average 1.33%
  - the lower the correlation, the larger the reduction in risk
- Diversification works only through $\rho < 1$
  - in a crash correlations rise, exactly when diversification is needed

Independence and Uncorrelatedness

- X and Y are **independent** if $F(x, y) = F_X(x)F_Y(y)$ for all x, y
  - the joint CDF is the product of the marginal CDFs
  - then $E[g(X)h(Y)] = E[g(X)]E[h(Y)]$ for **all** functions g, h (for example squares or absolute values)
- Independent $\Rightarrow$ uncorrelated ($g, h =$ identity); the converse is false
- Example: $X \sim N(0, 1)$, $Y = X^2$
  - $\text{Cov}(X, X^2) = E[X^3] = 0$: uncorrelated
  - but Y is a function of X: completely dependent
- Exception: for jointly Normal variables, uncorrelated $\Leftrightarrow$ independent

Returns Are Uncorrelated, Squared Returns Are Not



- Left: simulated $Y = X^2$, sample correlation -0.038, yet Y is determined by X
- Right: $\mathrm{Corr}(r_t, r_{t-1})$ is small: S&P 500 -0.099, DAX -0.017, Bitcoin -0.022
  - $\mathrm{Corr}(r_t^2, r_{t-1}^2)$ is far outside the band: 0.31, 0.15, 0.14; BET 0.33
  - band $\pm 1.96/\sqrt{n}$: the values compatible with zero correlation at the 5% level
  - the negative S&P 500 value comes mostly from crisis days
- BET: $\mathrm{Corr}(r_t, r_{t-1}) = 0.109$, an effect of thin trading

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Recap: Joint Distributions

- Covariance enters the variance of every portfolio
- Independent implies uncorrelated, not the other way round
- Returns: nearly uncorrelated, but their squares are correlated, so they are not independent

Conditional Expectation

- Conditional density: $f(y \mid x) = f(x, y)/f_X(x)$
  - the distribution of Y once we know that $X = x$: the joint density rescaled by the marginal one
- **Conditional expectation**: $E[Y \mid X = x] = \int y f(y \mid x)\, dy$
  - the mean of Y computed with the conditional density
  - $E[Y \mid X]$ is a random variable, a function of X
- It is the best forecast of Y from X: it minimises $E[(Y - g(X))^2]$ over all functions g
  - $E[(Y - g(X))^2]$: the mean squared error of the forecast $g(X)$
  - for returns: $E[r_t \mid \mathcal{F}_{t-1}]$, the forecast given all information $\mathcal{F}_{t-1}$ up to yesterday
- **Tower property** (law of iterated expectations): $E\big[E[Y \mid X]\big] = E[Y]$
  - averaging the conditional forecasts over all values of X gives back the unconditional mean

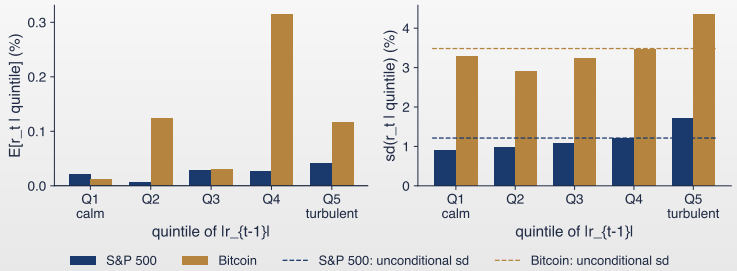
Conditional Variance and the Law of Total Variance

- **Conditional variance**: $\mathrm{Var}(Y \mid X) = E\left[(Y - E[Y \mid X])^2 \mid X\right]$
  - the spread of Y around its conditional mean, once X is known
- **Law of total variance**: $\mathrm{Var}(Y) = E\left[\mathrm{Var}(Y \mid X)\right] + \mathrm{Var}\left(E[Y \mid X]\right)$
  - $E[\mathrm{Var}(Y \mid X)]$: the average variance inside the groups defined by X (within)
  - $\mathrm{Var}(E[Y \mid X])$: how much the group means differ from each other (between)
  - total risk = within-group variance + between-group variance

Worked Example: Two Regimes

- X = the market regime, $Y = r$ = the daily return (in %)
  - calm: probability 0.7, mean 0.05%, standard deviation 0.8%
  - turbulent: probability 0.3, mean -0.10%, standard deviation 2%
- Mean (tower property): $E[r] = 0.7 \times 0.05 + 0.3 \times (-0.10) = 0.005\%$
- The two parts of the variance
  - within: $E[\mathrm{Var}(r \mid X)] = 0.7 \times 0.8^2 + 0.3 \times 2^2 = 1.648$
  - between: $\mathrm{Var}(E[r \mid X]) = 0.7(0.05 - E[r])^2 + 0.3(-0.10 - E[r])^2 = 0.0047$
- $\mathrm{Var}(r) = 1.653$, standard deviation 1.286%
  - almost all risk comes from the different variances of the regimes, not from their different means

Today's Return Given Yesterday's Move



- Quintiles of $|r_{t-1}|$: Q1 = the calmest 20% of yesterdays, Q5 = the most turbulent 20%
- Conditional mean: between 0.006% and 0.042% for the S&P 500, no usable pattern
- Conditional s.d.: S&P 500 from 0.91% (Q1) to 1.71% (Q5), ratio 1.88; Bitcoin ratio 1.33
- Law of total variance, S&P 500: 1.469 = 1.469 + 0.00013; the between part is 0.01%

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From Conditional Variance to GARCH

- Write the return as $r_t = \mu_t + \sigma_t z_t$, with z_t i.i.d., mean 0, variance 1
  - $\mu_t = E[r_t \mid \mathcal{F}_{t-1}]$: the conditional mean, close to constant
  - $\sigma_t^2 = \mathrm{Var}(r_t \mid \mathcal{F}_{t-1})$: the conditional variance, which changes with the news
- ARCH (autoregressive conditional heteroskedasticity) Engle (1982): $\sigma_t^2 = \omega + \alpha r_{t-1}^2$
  - today's variance is a base level plus a share of yesterday's squared return
  - $\omega > 0$: the base level; $\alpha \geq 0$: how strongly a large move yesterday raises today's variance
- GARCH Bollerslev (1986): $\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2$ (Chapter 9)
  - $\beta \geq 0$: the persistence; a large β keeps a high variance high for many days
- Exactly the pattern in the chart: a large $|r_{t-1}|$ raises today's conditional s.d.
- The unconditional distribution is a mixture over σ_t
  - it has heavier tails than the Normal distribution, even if z_t is Normal

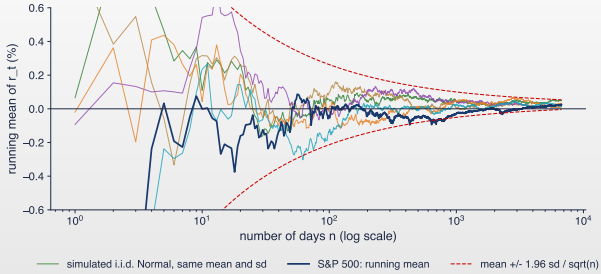
Recap: Conditioning

- $E[Y \mid X]$ is the best forecast; the tower property averages it back to $E[Y]$
- Returns: conditional mean almost flat, conditional variance strongly time-varying
- Modelling σ_t^2 is the job of ARCH and GARCH

Two Limit Theorems

- **LLN** (law of large numbers): for i.i.d. X_i with $E|X| < \infty$, $\bar{X}_n \to \mu$ as $n \to \infty$
  - $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$: the sample mean; $\mu = E[X]$: the true mean
  - the reason why averages, frequencies and Monte Carlo estimates work
- **CLT** (central limit theorem): with finite variance, $\sqrt{n}(\bar{X}_n - \mu)/\sigma \to N(0,1)$ (Chapter 2)
  - the standardised sample mean has, for large n, approximately the standard Normal distribution
  - gives the speed: the error of $\bar{X}_n$ is about $\sigma/\sqrt{n}$
  - infinite variance: sums converge to α-stable laws instead (Chapter 3)
- Both need some form of independence; dependent returns converge more slowly

The LLN for the Mean Return: Slow



- S&P 500, 26.6 years: mean 0.0247% a day, standard error $\sigma/\sqrt{n} = 0.0148\%$
  - t = mean / standard error $= 1.67$; $|t| > 1.96$ would mean a mean significantly different from 0 at 5%
- Annual mean 6.2%, 95% interval $[-1.1\%, 13.5\%]$
  - even 26 years do not pin down the expected return
  - for $t = 3$ at this mean and volatility we would need about 86 years of data

SFM_ch4_monte_carlo

The Monte Carlo Idea

- Estimate $\theta = E[g(X)]$ by $\hat{\theta}_N = \frac{1}{N} \sum_{i=1}^{N} g(X_i)$ with simulated X_i
  - θ: the unknown quantity; g: a function of the simulated variable; N: the number of simulations
  - a probability is an expectation: $P(A) = E[1_A]$, with $1_A = 1$ if A occurs and 0 otherwise
  - the LLN makes $\hat{\theta}_N \to \theta$; the CLT gives the error $\text{sd}(g(X))/\sqrt{N}$
- Named after the Monaco casino by Ulam and von Neumann, Los Alamos, 1940s Metropolis & Ulam (1949)
- In finance Glasserman (2003)
  - prices of path-dependent options, portfolio VaR (Chapter 10), stress scenarios
  - any expectation without a closed form
- Two ingredients: uniform random numbers and a way to turn them into any distribution

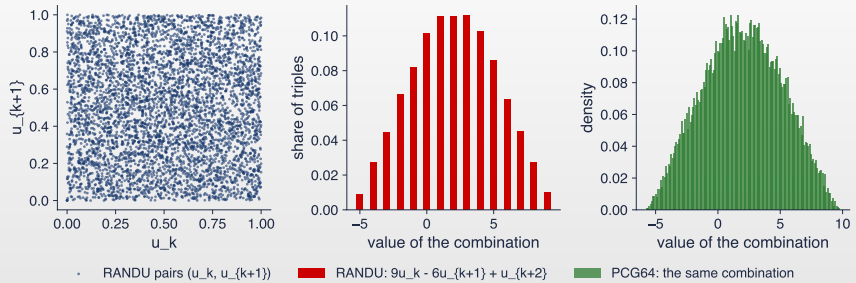
Pseudo-Random Numbers: Linear Congruential Generators

- **LCG** (linear congruential generator): $x_{k+1} = (ax_k + c) \bmod M$, $u_k = x_k/M \in [0, 1)$
  - each integer x_{k+1} is computed from the previous one x_k
  - a: the multiplier; c: the increment; M: the modulus; $\bmod M$: the remainder of the division by M
  - u_k: the uniform number in $[0, 1)$ obtained by dividing by M
- Deterministic: the same seed x_0 gives the same sequence
  - this makes results reproducible
  - the **period**: the number of values before the sequence repeats; at most M
- Real generators use huge M; the period must be far longer than any simulation

Worked Example: an LCG with $M = 16$

- SFErangen1: $a = 5$, $c = 1$, $M = 16$, $x_0 = 7$
  - first step: $x_1 = (5 \times 7 + 1) \bmod 16 = 36 \bmod 16 = 4$
  - the sequence x: 7, 4, 5, 10, 3, 0, 1, 6, 15, 12, 13, 2, 11, 8, 9, 14, 7
- Period 16, the maximum possible, M
  - with $a = 4$ the sequence sticks after two steps (period 1)
- Good uniform marginals are not enough
  - consecutive numbers must also look independent: the next chart

RANDU: Random Numbers Fall Mainly in the Planes (SFErandu)



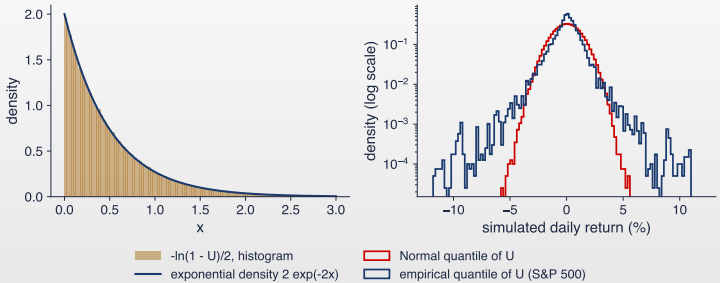
- RANDU, IBM, 1960s: $x_{k+1} = 65539\, x_k \bmod 2^{31}$; pairs look uniform (left), correlation 0.003
- But $9u_k - 6u_{k+1} + u_{k+2}$ takes only 15 integer values, from -5 to 9: all triples lie on 15 planes Marsaglia (1968)
- Modern generators: Mersenne Twister Matsumoto & Nishimura (1998), PCG64 (the NumPy default, right panel)

SFM_ch4_random_numbers

The Inverse Transform Method

- **Theorem**: if $U \sim U(0,1)$ and F is a CDF, then $X = F^{-1}(U)$ has CDF F
  - $U(0,1)$: the uniform distribution on $[0,1]$, so $P(U \leq u) = u$
  - in words: a uniform number, read as a probability, is turned into the quantile of that order
  - proof: $P(X \leq x) = P(F^{-1}(U) \leq x) = P(U \leq F(x)) = F(x)$
- Exponential with rate $\lambda > 0$: $F(x) = 1 - e^{-\lambda x}$, so $X = -\ln(1-U)/\lambda$
  - obtained by solving $u = F(x)$ for x; the mean is $1/\lambda$
  - $\lambda = 2$, $u = 0.9$: $x = -\ln(0.1)/2 = 1.151$
- Normal: $X = \mu + \sigma\,\Phi^{-1}(U)$
  - Φ^{-1}: the quantile function of $N(0,1)$; μ, σ: the desired mean and standard deviation
- Data: use the empirical quantile function $\hat{F}^{-1}$
  - this is resampling past days: historical simulation and the bootstrap

Inverse Transform: Exponential, Normal and Historical



- Left: 200 000 draws $-\ln(1-U)/2$ match the exponential density; sample mean 0.500 (theory $1/\lambda = 0.5$)
- Right: the same uniforms through the Normal and the empirical S&P 500 quantile functions
- Draws below -5%: 0.0035% Normal (7 draws; exact 0.0017%, the gap is simulation noise) vs 0.34% historical (data: 0.34%); worst draw -5.7% vs -12.7%

SFM_ch4_random_numbers

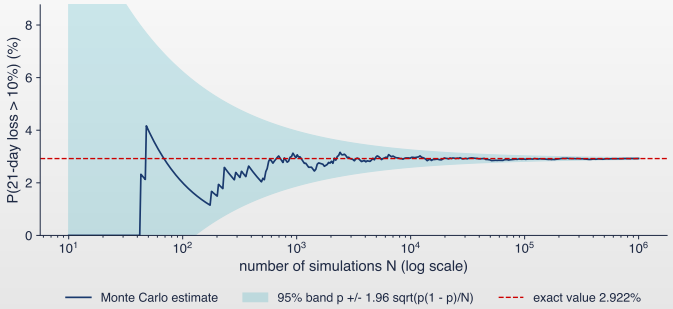
Worked Example: a Monthly Loss Probability by Monte Carlo (1/2)

- Model: S&P 500 daily log returns i.i.d. $N(0.0247, 1.213^2)$, in %
  - $N(m, s^2)$: the Normal distribution with mean m and variance s^2
- Event: the 21-day log return (one month) is below -10%
  - p: its probability, the quantity we want
- Exact: the sum of 21 i.i.d. Normal returns is $N(0.519, 5.56^2)$
  - mean 21×0.0247, standard deviation $1.213\sqrt{21}$
  - standardise: $z = (-10 - 0.519)/5.56 = -1.892$
  - $p = \Phi(z) = 2.922\%$, with Φ the CDF of $N(0, 1)$

Worked Example: a Monthly Loss Probability by Monte Carlo (2/2)

- Monte Carlo: simulate N months; $\hat{p}_N$ = share of simulated months below -10%
  - standard error $\sqrt{p(1-p)/N}$: the typical distance between $\hat{p}_N$ and p
- Results
  - $N = 1\,000$: 3.000% (s.e. 0.533%)
  - $N = 100\,000$: 2.901% (s.e. 0.053%)
  - 10 times more precision needs 100 times more simulations
- For a relative error of 10% at the 95% level: $N \approx 1.96^2(1-p)/(0.1^2 p) = 12\,762$
  - obtained from $1.96 \times \sqrt{p(1-p)/N} = 0.1\,p$: the half-width of the interval is 10% of p
  - a rare event needs many simulations

Monte Carlo Converges at Rate $1/\sqrt{N}$



- The estimate wanders for small N and settles inside the 95% band, which shrinks like $1/\sqrt{N}$
- $N = 10^6$: 2.926% vs exact 2.922%, s.e. 0.017%
- Monte Carlo removes the simulation error, not the model error: Seminar 4 (B3) repeats it with resampled days

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Recap: Random Numbers and Monte Carlo

- Pseudo-random generators are deterministic; check them in several dimensions
- $F^{-1}(U)$ turns uniforms into any distribution, including the empirical one
- Monte Carlo error $\propto 1/\sqrt{N}$; always report it

The Binomial Process

- Each period the price is multiplied by u (up) with probability p, or by $d < u$ (down) with probability $1 - p$
  - u, d: the up and down factors (e.g. $u = 1.2$: $+20\%$); the periods are independent
- After n steps with K up moves: $S_n = S_0 u^K d^{n-K}$
  - S_0: the initial price; $K \sim B(n, p)$: the number of up moves is binomial
  - $\ln S_n = \ln S_0 + K \ln u + (n - K) \ln d$: a random walk in log prices (Section 9)
- $E[S_n] = S_0 (pu + (1 - p)d)^n$, because the steps are independent
  - $pu + (1 - p)d$: the expected growth factor of one step
  - the basis of the binomial option pricing model Cox, Ross & Rubinstein (1979); textbook: Franke, Härdle & Hafner (2019), Ch. 7

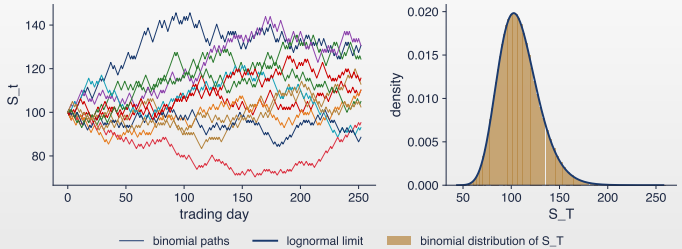
Worked Example: a Two-Step Tree

- $S_0 = 100$, $u = 1.2$, $d = 0.8$, $p = 0.6$
- Terminal values and probabilities
  - uu: 144 with $p^2 = 0.36$; ud or du: 96 with $2p(1-p) = 0.48$; dd: 64 with $(1-p)^2 = 0.16$
- $E[S_2] = 100 \times (0.6 \times 1.2 + 0.4 \times 0.8)^2 = 100 \times 1.04^2 = 108.16$
- $P(S_2 < 100) = 0.48 + 0.16 = 0.64$
  - one up and one down move lose money, because $ud = 0.96 < 1$
- With $p^* = (1-d)/(u-d) = 0.50$ the price is a fair game: $E[S_{t+1} \mid S_t] = S_t$ (Section 9)
  - p^* solves $p^* u + (1-p^*)d = 1$: the expected growth factor of a step equals 1

Calibrating the Tree to a Market (Cox–Ross–Rubinstein)

- **CRR** (Cox–Ross–Rubinstein) with step Δt: $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$, $p = (e^{\mu \Delta t} - d)/(u - d)$
  - Δt: the length of one step, in years; μ: the annual drift; σ: the annual volatility
  - $d = 1/u$: an up move followed by a down move returns to the start
  - chosen so that the mean and the variance of each step match a market with drift μ and volatility σ
- S&P 500, 2000–2026: $\sigma = 19.26\%$, $\mu = 8.08\%$ a year; daily steps, $\Delta t = 1/252$
  - $u = 1.0122$, $d = 0.9879$, $p = 0.5102$
- As $\Delta t \to 0$, $\ln S_T$ becomes Normal (CLT for K)
  - T: the horizon; the tree converges to GBM (Section 10)

Binomial Paths and the Lognormal Limit (SFEBinomp)



- Left: 12 paths over one year of daily steps, $S_0 = 100$
- Right: distribution of S_T after 252 steps ($T = 1$ year); $E[S_T] = 108.42 = 100e^{\mu}$
  - 5% quantile 78.5 vs 77.5 in the lognormal limit
- $P(S_T < 100)$: 34.9% binomial ($P(S_T \leq 100) = 39.7\%$) vs 37.3% lognormal
  - the exact value $S_T = 100$ has probability 4.8% in the tree; the lognormal value lies between the two

SFM_ch4_binomial

Recap: The Binomial Model

- Up or down each period: $S_n = S_0 u^K d^{n-K}$ with binomial K
- CRR matches μ and σ; the limit is lognormal
- A special p^* turns the price into a fair game: the key to option pricing

Stochastic Processes and Information

- A **stochastic process** $\{X_t\}$: a family of random variables indexed by time
  - one realisation = one path; a price series is a single path of the process
- $\mathcal{F}_t$: the information available at time t (past prices, news)
  - it grows with t: nothing is forgotten (a **filtration**)
- **(Weakly) stationary**: $E[X_t]$ and $\mathrm{Cov}(X_t, X_{t+h})$ do not depend on t
  - h: the lag between two dates; the mean and the dependence look the same at every date
  - returns: roughly yes; prices: no
- Textbook: Franke, Härdle & Hafner (2019), Ch. 4 and Sec. 11.3

White Noise and the Random Walk

- **White noise**: $E[\varepsilon_t] = 0$, $\text{Var}(\varepsilon_t) = \sigma^2$, $\text{Cov}(\varepsilon_t, \varepsilon_s) = 0$ for $t \neq s$
  - ε_t: the shock at time t; mean zero, constant variance σ^2, no correlation across dates
  - i.i.d. noise is stronger: the ε_t are independent, not only uncorrelated
- **Random walk**: $S_t = S_{t-1} + \mu + \varepsilon_t = S_0 + \mu t + \sum_{s=1}^{t} \varepsilon_s$
  - each step adds the drift μ and a new shock; S_0: the starting value
  - $E[S_t] = S_0 + \mu t$; $\text{Var}(S_t) = t\sigma^2$ grows with t: not stationary
  - simulated with $\sigma = 1$: variance 10.1, 49.8, 100.9 at $t = 10, 50, 100$
- Log prices as a random walk = returns as white noise
  - the benchmark of Chapter 7 (efficient markets)

Martingales

- $\{X_t\}$ is a **martingale** if $E|X_t| < \infty$ and $E[X_{t+1} \mid \mathcal{F}_t] = X_t$
  - $E|X_t| < \infty$: the mean exists; $\mathcal{F}_t$: the information up to t
  - a fair game: the best forecast of tomorrow is today
- A **martingale difference**: $\varepsilon_t = X_t - X_{t-1}$ with $E[\varepsilon_t \mid \mathcal{F}_{t-1}] = 0$
  - the change cannot be predicted from the past
- Examples
  - a random walk without drift; $S_t^2 - t\sigma^2$ for the same walk; the binomial price under p^*
- Martingale $\neq$ i.i.d. increments: the conditional **variance** may change, as in GARCH
  - the efficient-market idea in its weak form Fama (1970): excess returns (above the risk-free rate) are a martingale difference

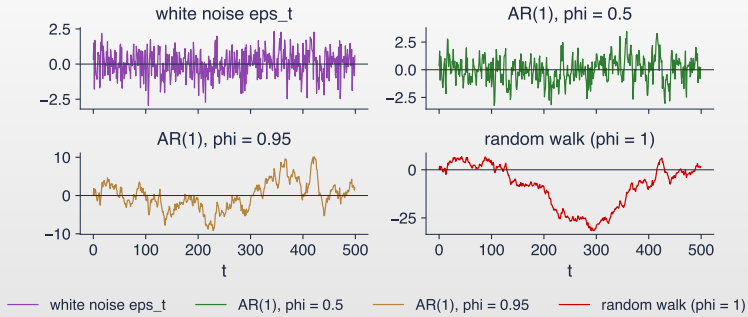
The AR(1) Process

- **AR(1)** (autoregressive of order 1): $X_t = c + \phi X_{t-1} + \varepsilon_t$, ε_t white noise
  - today's value is a constant, plus a share ϕ of yesterday's value, plus a new shock
  - c: the constant; ϕ: the persistence; σ^2: the variance of ε_t
  - $\phi = 0$: white noise; $\phi = 1$, $c = \mu$: random walk
- Stationary iff $|\phi| < 1$; then
  - mean $c/(1-\phi)$, variance $\sigma^2/(1-\phi^2)$
  - autocorrelation at lag h: $\rho(h) = \mathrm{Corr}(X_t, X_{t-h}) = \phi^h$, decaying geometrically
  - half-life of a shock: $\ln 0.5 / \ln \phi$ periods, the time until half of a shock is gone
- Uses
  - interest rates and spreads (mean reversion)
  - the squared returns behave like a persistent AR, which leads to GARCH

The AR(1) Process: Simulation

- Simulated with $\sigma = 1$, $\phi = 0.5$
  - variance 1.34 (theory $1/(1 - 0.5^2) = 1.33$)
  - $\rho(5) = 0.034$ (theory $0.5^5 = 0.031$); half-life 1.0 periods
- Simulated with $\sigma = 1$, $\phi = 0.95$
  - variance 10.20 (theory 10.26)
  - $\rho(5) = 0.772$; half-life 13.5 periods
- The closer ϕ is to 1, the larger the variance and the slower a shock fades

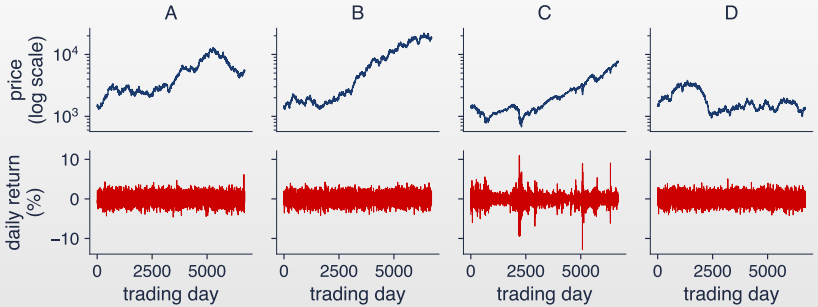
White Noise, AR(1) and Random Walk, Same Shocks



- The larger ϕ, the longer a shock lives: the AR(1) with $\phi = 0.95$ wanders, but returns to 0
- The random walk never returns on purpose: shocks are permanent and its variance grows with t

SFM_ch4_processes

Question for the Room: Which One Is Real?



- One column is the S&P 500, 2000–2026; three are GBM paths (i.i.d. Normal returns) with the same mean and volatility
- **What do you think?** Which column is real, from the prices alone?
- **What do you think?** Which column is real, from the returns?

SFM_ch4_processes

Answer: Column C

- **Answer**: column C is the S&P 500
- Prices: hard to tell; random walks with drift produce booms and long falls too
  - this is why Bachelier (1900) and Osborne (1959) modelled prices as random walks
- Returns: obvious; the real series has quiet years and violent clusters (2008, 2020)
  - excess kurtosis 10.7 vs at most 0.02 in the simulations; $\mathrm{Corr}(r_t^2, r_{t-1}^2) = 0.31$ vs at most 0.006
  - largest absolute daily move 12.8% vs 6.2%
- Lesson: a model can fit the price path and still be wrong about risk

Recap: Discrete-Time Processes

- White noise: uncorrelated; i.i.d. noise: independent; a martingale difference: unpredictable mean
- Random walk: permanent shocks, variance $t\sigma^2$; AR(1) with $|\phi| < 1$: shocks fade at rate ϕ^h
- Prices look like random walks; returns are not i.i.d.

From Pollen to Prices: Brownian Motion

- 1827: the botanist Robert Brown sees pollen particles jiggling in water
- 1900: Louis Bachelier models the Paris bond market with the same motion, five years before physics Bachelier (1900)
- 1905: Albert Einstein explains it by molecular collisions Einstein (1905)
- 1923: Norbert Wiener builds the rigorous mathematical object Wiener (1923)
- 1959: Osborne (1959) applies it to log prices: geometric Brownian motion



Robert Brown (1773–1858)

Photo: Maull & Polyblank (1855); public domain; Wikimedia Commons

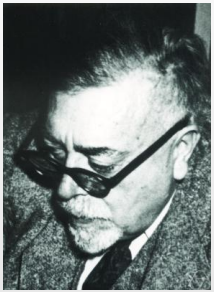


Louis Bachelier (1870–1946)

Photo: unknown author; public domain; Wikimedia Commons

The Wiener Process

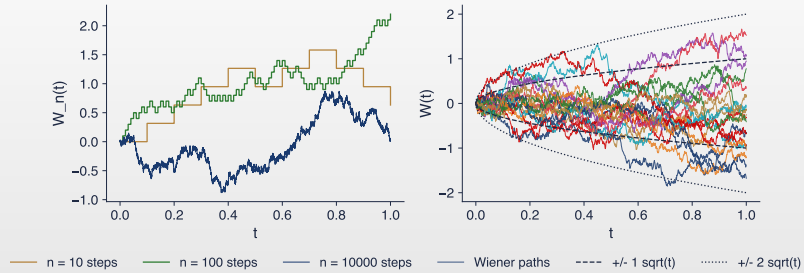
- $\{W(t), t \geq 0\}$ is a **Wiener process** (standard Brownian motion) if
  - $W(0) = 0$
  - increments over disjoint intervals are independent
  - $W(t) - W(s) \sim N(0, t - s)$ for $s < t$: the variance equals the elapsed time
  - the paths are continuous
- Limit of the scaled random walk $W_n(t) = (\varepsilon_1 + \cdots + \varepsilon_{\lfloor nt \rfloor})/\sqrt{n}$ (Donsker)
  - $\varepsilon_i = \pm 1$ with probability $1/2$ each; n steps per unit of time
  - $\lfloor nt \rfloor$: the number of steps up to time t (integer part); dividing by $\sqrt{n}$ keeps the variance equal to t
- Textbook: Franke, Härdle & Hafner (2019), Ch. 5



Norbert Wiener (1894–1964)

Photo: Konrad Jacobs; CC BY-SA 2.0 DE; Wikimedia Commons

From Random Walks to the Wiener Process (SFEWienerProcess)



- Left: scaled random walks with 10, 100 and 10 000 steps; the steps vanish, the randomness stays
- Right: 20 Wiener paths; $\mathrm{Var}(W(t)) = t$, so they spread like $\pm\sqrt{t}$
  - simulated $\mathrm{Var}(W(1)) = 1.006$; 68.2% of $W(1)$ inside ± 1 (theory 68.3%)

SFM_ch4_wiener_gbm

Strange Properties of Wiener Paths

- Continuous everywhere, differentiable nowhere
  - the slope $(W(t+h) - W(t))/h \sim N(0, 1/h)$ has a variance $1/h$ that grows without bound as $h \to 0$
- **Quadratic variation**: $\sum_k (W(t_{k+1}) - W(t_k))^2 \to t$ on a finer and finer grid
  - $0 = t_0 < t_1 < \cdots < t$: the grid; the sum of squared increments tends to the elapsed time t
  - 1000 steps on $[0, 1]$: mean 0.999, s.d. 0.045 across 10 000 paths
  - the reason why $(dW)^2 = dt$ in Itô calculus, and why realised variance measures volatility (Chapter 8)
- $\mathrm{Cov}(W(s), W(t)) = \min(s, t)$: simulated $\mathrm{Cov}(W(0.5), W(1)) = 0.496$
- W is a martingale; so is $W(t)^2 - t$

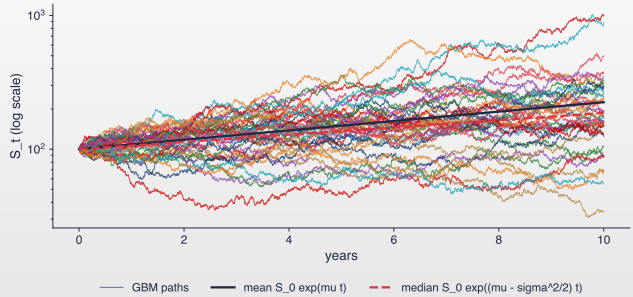
Geometric Brownian Motion

- **GBM** (geometric Brownian motion): $dS_t = \mu S_t\, dt + \sigma S_t\, dW_t$
  - dS_t: the change of the price over a very short interval dt
  - dW_t: the increment of the Wiener process over dt, distributed $N(0, dt)$
  - the relative change dS/S has drift μ and volatility σ, both per year
- Solution: $S_t = S_0 \exp\left((\mu - \sigma^2/2)t + \sigma W_t\right)$
  - $\ln(S_t/S_0) \sim N\left((\mu - \sigma^2/2)t, \sigma^2 t\right)$
  - lognormal prices, i.i.d. Normal log returns
- The model behind Black & Scholes (1973) (textbook: Franke, Härdle & Hafner (2019), Ch. 6)

GBM: Mean, Median and Simulation

- Mean $E[S_t] = S_0 e^{\mu t}$; median $S_0 e^{(\mu - \sigma^2/2)t}$
  - the median is lower: half of the paths end below $S_0 e^{(\mu - \sigma^2/2)t}$
  - the gap, $\sigma^2/2$ a year in log terms, is the volatility drag
- Exact simulation on a grid: $S_{t+\Delta t} = S_t \exp\left((\mu - \sigma^2/2)\Delta t + \sigma\sqrt{\Delta t}\, Z\right)$
  - Δt: the grid step, in years; $Z \sim N(0,1)$, a new independent draw at each step
  - exact because each log increment is Normal with the right mean and variance

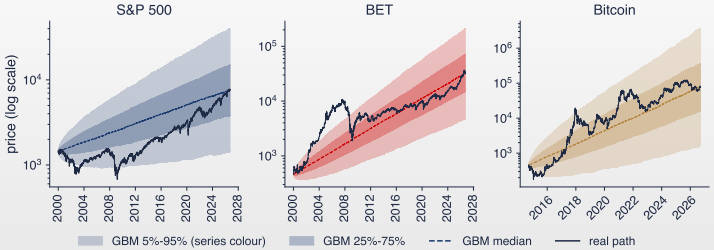
GBM Calibrated to the S&P 500 (SFEsimGBM)



- Calibration, 2000–2026: $\sigma = 19.26\%$, log drift $\mu - \sigma^2/2 = 6.23\%$, so $\mu = 8.08\%$ a year
- After $T = 10$ years: mean 224.4, median 186.4
  - 62.0% of the paths end below the mean; theory: $\Phi(\sigma\sqrt{T}/2) = 62.0\%$, with Φ the $N(0,1)$ CDF
- Probability of a loss after 10 years: 15.3%

SFM_ch4_wiener_gbm

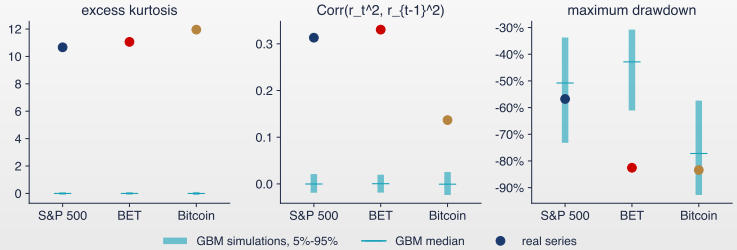
Real Paths in the GBM Fan



- Each fan: 4000 GBM paths with the drift and volatility of the series itself, started on its first day
- S&P 500: the path reached the 0.5% quantile on 23 July 2002, then the median at the end
  - outside the 90% band 7% of the time
- BET: outside the band 24% of the time
  - the 2000–2007 boom and the 2008 crash are not GBM-like
- Bitcoin: $\sigma = 66.6\%$ a year makes the fan enormous
  - annualised with $A = 365$: crypto trades every calendar day ($A = 252$ for the indices)

SFM_ch4_wiener_gbm

What GBM Misses



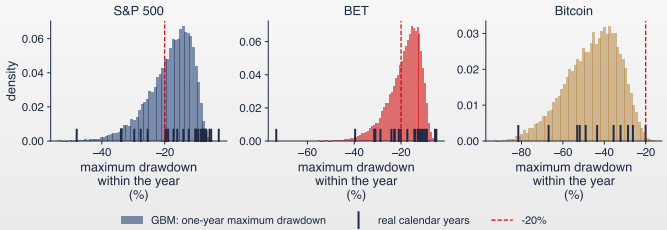
- Excess kurtosis: S&P 500 10.7, BET 11.1, Bitcoin 12.0; 90% of 300 GBM simulations lie within $[-0.10, 0.09]$
- Volatility clustering: $\text{Corr}(r_t^2, r_{t-1}^2) = 0.31, 0.33, 0.14$ vs at most 0.03 under GBM
- Maximum drawdown $\min_t \left(S_t / \max_{s \le t} S_s - 1 \right)$: the largest fall from a previous peak
  - S&P 500 -57% is ordinary for GBM; BET -83% is beyond all but 5% of the paths (5% quantile -61%)
- 4-sigma days: 46 (S&P 500) vs at most 1 in 95% of the GBM paths; crypto has the same problem Pele et al. (2023)

SFM_ch4_wiener_gbm

Recap: Wiener Process and GBM

- Wiener process: independent Normal increments with variance equal to the elapsed time
- GBM: lognormal prices, median below mean, the benchmark of option pricing
- Real returns have heavy tails and volatility clusters that GBM cannot produce

An Open Question: Does GBM Get Drawdown Risk Right?



- Drawdown (What GBM Misses), here within a calendar year
  - beyond 20%: S&P 500 in 6 of 26 years (23%) vs 34% under GBM; BET 46% vs 34%
- Beyond 40%: GBM expects 0.14 such years for the S&P 500 and 0.18 for the BET
  - both had 1 (2008: −48% and −73%)
- Open question
  - does GBM overstate moderate drawdowns and understate extreme ones because calm and crisis years cluster?

SFM_ch4_drawdown_open

How AI Could Help

- **Literature**: find studies of drawdown distributions and of their link to volatility clustering, and summarise them
- **Code**: draft simulations of yearly drawdowns under GBM, the i.i.d. bootstrap and a GARCH model
- **Design**: propose a test that compares the observed number of bad years with the model probability
- Example prompt
  - `Write Python code that simulates 10,000 one-year paths of daily S&P 500 returns under GBM and under an i.i.d. bootstrap of 2000-2026 returns, and returns the probability of a drawdown beyond 20% and 40% within the year.`

What to Check

- Definitions: drawdown from the running peak within the year, not from the first day of the year
- Calibration: the same μ and σ, the same number of trading days a year as the data
- Inference: 26 years is a small sample; compute a binomial p-value, not only shares
- Mechanism: the i.i.d. bootstrap keeps heavy tails but removes clustering; the difference isolates the effect of clustering
- References: every cited paper must exist; check the DOI

Project Seed

- **Question**: how wrong is GBM about the probability of a bad year, and why?
  - data: S&P 500, DAX, BET and Bitcoin from the course data
- Steps
  - compute within-year maximum drawdowns for every calendar year
  - simulate the same statistic under GBM, the i.i.d. bootstrap and a GARCH(1,1) model (Chapter 9)
  - compare the shares beyond 20% and 40% with binomial tests
- Deliverable: one table, one chart, and a paragraph on what the data can and cannot show
- Declare any AI use, and list the errors of the AI that you corrected Wang et al. (2023)

Key Formulas

Quantity	Formula
Variance of a sum	$\mathrm{Var}(aX+bY) = a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\,\mathrm{Cov}(X,Y)$
Correlation	$\rho = \mathrm{Cov}(X,Y)/(\sigma_X\sigma_Y)$
Tower property	$E[E[Y \mid X]] = E[Y]$
Total variance	$\mathrm{Var}(Y) = E[\mathrm{Var}(Y \mid X)] + \mathrm{Var}(E[Y \mid X])$
Inverse transform	$X = F^{-1}(U)$, $U \sim U(0,1)$
Monte Carlo error	$\sqrt{p(1-p)/N}$
Random walk	$S_t = S_{t-1} + \mu + \varepsilon_t$, $\mathrm{Var}(S_t) = t\sigma^2$
AR(1)	$\mathrm{Var}(X_t) = \sigma^2/(1-\phi^2)$, $\rho(h) = \phi^h$
GBM	$S_t = S_0 \exp((\mu - \sigma^2/2)t + \sigma W_t)$, $E[S_t] = S_0 e^{\mu t}$

Check Yourself

- **Question**: $\sigma_1 = 2\%$, $\sigma_2 = 3\%$, $\rho = 0.5$; what is the volatility of the 50/50 portfolio?
  - **Answer**: $\sqrt{0.25 \times 4 + 0.25 \times 9 + 2 \times 0.25 \times 0.5 \times 2 \times 3} = \sqrt{4.75} = 2.18\%$
- **Question**: GBM with $\mu = 8\%$, $\sigma = 20\%$, $S_0 = 100$; what are the mean and the median of S_1?
  - **Answer**: mean $100e^{0.08} = 108.33$, median $100e^{0.06} = 106.18$
- **Question**: AR(1) with $\phi = 0.8$ and $\sigma = 1$; what is its variance?
  - **Answer**: $1/(1 - 0.64) = 2.78$
- Next: Chapter 5, heavy tails and extreme value theory

References (1/2)

- Bachelier, L. (1900). Théorie de la spéculation. *Annales scientifiques de l'École Normale Supérieure*, 17, 21–86.
- Black, F., Scholes, M. (1973). The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3), 637–654.
- Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327.
- Borak, S., Härdle, W.K., López-Cabrera, B. (2013). *Statistics of Financial Markets: Exercises and Solutions*, 2nd ed. Springer.
- Cont, R. (2001). Empirical properties of asset returns: stylized facts and statistical issues. *Quantitative Finance*, 1(2), 223–236.
- Cox, J.C., Ross, S.A., Rubinstein, M. (1979). Option pricing: a simplified approach. *Journal of Financial Economics*, 7(3), 229–263.
- Einstein, A. (1905). Über die von der molekularkinetischen Theorie der Wärme geforderte Bewegung von in ruhenden Flüssigkeiten suspendierten Teilchen. *Annalen der Physik*, 322(8), 549–560.
- Engle, R.F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1007.
- Fama, E.F. (1970). Efficient capital markets: a review of theory and empirical work. *Journal of Finance*, 25(2), 383–417.
- Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.

References (2/2)

- Glasserman, P. (2003). *Monte Carlo Methods in Financial Engineering*. Springer.
- Kolmogoroff, A. (1933). *Grundbegriffe der Wahrscheinlichkeitsrechnung*. Springer, Berlin.
- Marsaglia, G. (1968). Random numbers fall mainly in the planes. *Proceedings of the National Academy of Sciences*, 61(1), 25–28.
- Matsumoto, M., Nishimura, T. (1998). Mersenne twister: a 623-dimensionally equidistributed uniform pseudo-random number generator. *ACM Transactions on Modeling and Computer Simulation*, 8(1), 3–30.
- Metropolis, N., Ulam, S. (1949). The Monte Carlo method. *Journal of the American Statistical Association*, 44(247), 335–341.
- Osborne, M.F.M. (1959). Brownian motion in the stock market. *Operations Research*, 7(2), 145–173.
- Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). Are cryptos becoming alternative assets? *European Journal of Finance*, 29(10), 1064–1105.
- Wang, H., Fu, T., Du, Y., Gao, W., et al. (2023). Scientific discovery in the age of artificial intelligence. *Nature*, 620, 47–60.
- Wiener, N. (1923). Differential-space. *Journal of Mathematics and Physics*, 2(1–4), 131–174.